# Project Report: Economic Intelligence Assistant

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**Institution:** Texas A&M University

**Platform:** Snowflake AI Data Cloud (Cortex & Streamlit)

## 1. Executive Summary

The **Economic Intelligence Assistant** is a Retrieval-Augmented Generation (RAG) application designed to provide grounded, cited insights into US banking and demographic data. By integrating Snowflake's **Cortex Search** and **Arctic** embedding models, the project solves the challenge of LLM hallucinations in financial contexts, ensuring every answer is backed by verifiable Marketplace data.

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## 2. Data Architecture & Sources

The project utilizes high-fidelity datasets sourced from the **Snowflake Marketplace** to ensure institutional-grade accuracy:

- **Banking Analytics Bundle (FDIC):** Comprehensive data on insured financial institutions, assets, and regulatory history.
- **Demographics Data Bundle (InSights):** Granular US population metrics, including income distribution, poverty levels, and educational attainment.

**Schema Overview:**
Both datasets were structured into normalized relational tables within Snowflake. The banking dataset includes attributes such as institution identifiers, asset size, regulatory status, and geographic indicators. The demographics dataset contains region-level attributes including median income, population distribution, education levels, and poverty rates.

To support retrieval, all textual fields (e.g., descriptions, summaries, and categorical annotations) were consolidated and transformed into a unified schema for downstream chunking and embedding.

**Representative Fields:**

- *Banking Dataset:* `institution_name`, `total_assets`, `deposit_amount`, `regulatory_status`, `state`
- *Demographics Dataset:* `median_income`, `population`, `education_level`, `poverty_rate`, `region`

These fields provide both quantitative and descriptive context, enabling the system to answer analytical as well as explanatory queries.

**Cross-Dataset Complementarity:**
The combination of banking and demographic datasets enables richer economic reasoning beyond isolated analysis. While the banking dataset captures institutional performance and financial health, the demographics dataset provides socio-economic context such as income distribution and population characteristics.

Together, they allow the assistant to answer higher-level questions. For example, how regional income levels may influence credit risk, or how population trends correlate with banking activity, thereby enhancing the system's analytical depth.

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# 3. Technical Implementation

## Step 1: Data Ingestion & Semantic Chunking

Raw data was processed using **Snowpark Python**. To optimize retrieval, we implemented recursive character splitting to break dense regulatory documents into manageable fragments.

- **Chunk Size:** 512 characters.
- **Overlap:** 50 characters (to preserve semantic context).
- **Output:** `CHUNKED_DOCUMENTS` table containing ~36,880 searchable entries.

## Step 2: Vector Embedding & Cortex Search

We deployed a semantic search layer to move beyond simple keyword matching:

- **Embedding Model:** `snowflake-arctic-embed-m-v1.5` (State-of-the-art for retrieval).
- **Service: Cortex Search Service** (`ECONOMIC_SEARCH`), configured for incremental refreshes.
- **Infrastructure:** A dedicated `RAG_SEARCH_WH` warehouse was created to isolate search workloads from ingestion logic.

## Step 3: The RAG Logic (Generation)

The "Brain" of the assistant uses **Llama 3.1 70B** via the `SNOWFLAKE.CORTEX.COMPLETE` function.

- **Prompt Engineering:** The system is instructed to act as a strict researcher, citing sources using **[Source N]** notation.
- **Guardrails:** Explicit instructions prevent the model from using external knowledge, forcing a "Data does not cover this" response if context is missing.

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# 4. User Interface

The application is deployed via **Streamlit in Snowflake (SiS)**, offering a low-latency, secure interface for end-users.

- **Features:** Real-time chat, dataset filtering, adjustable chunk retrieval (k-value), and a dynamic "Source Citations" sidebar.
- **Security:** The app runs entirely within the Snowflake perimeter, ensuring data privacy and governance.

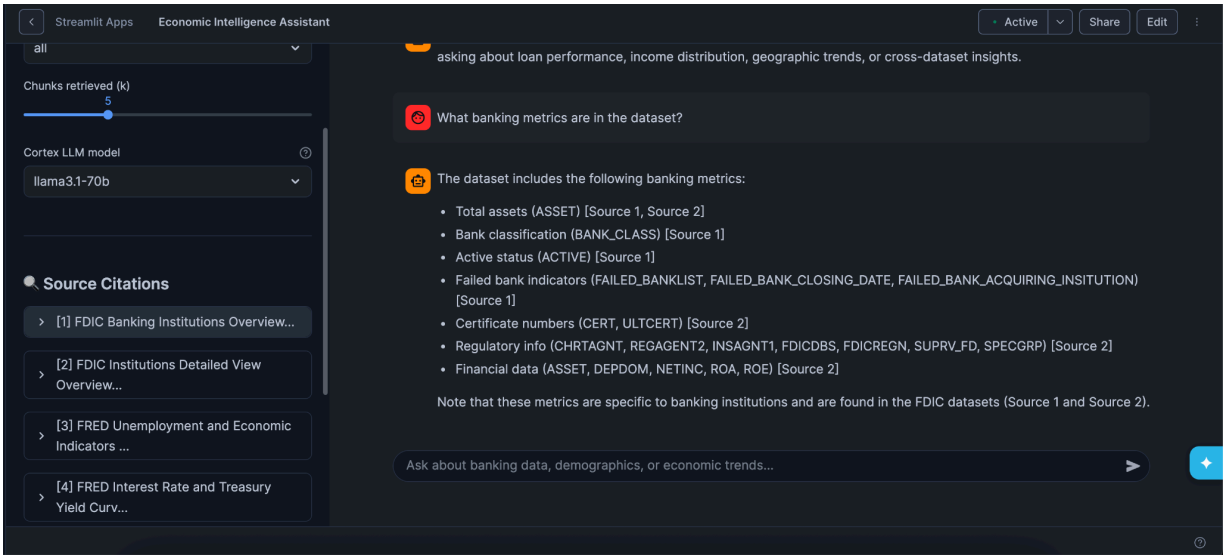


*Figure: Streamlit-based Economic Intelligence Assistant interface showing chat interaction, dataset filtering, and source citation panel.*

**User Interaction Flow:**

1. The user enters a natural language query in the chat input or selects a suggested prompt.
2. The system retrieves the top-k most relevant document chunks from Cortex Search based on semantic similarity.
3. Retrieved context is passed to the LLM (via COMPLETE()), which generates a grounded response with inline citations.
4. The response is displayed in the chat window, while supporting source snippets are shown in the “Source Citations” sidebar.
5. Users can refine results by adjusting the dataset filter or the number of retrieved chunks (k-value).

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## 5. Performance Evaluation

To validate the system, we ran an automated **LLM-as-a-Judge** evaluation suite using 20 "Ground Truth" Q&A pairs.

| Metric | Score | Definition |
|---|---|---|
| **Avg Precision** | **94.0%** | The search engine's ability to find the relevant data chunk. |
| **Avg Faithfulness** | **82.5%** | The LLM's adherence to the retrieved context without hallucinating. |

**Evaluation Methodology:**
Precision was computed by measuring the proportion of retrieved chunks that were relevant to the ground-truth answer for each query (Precision@k). Faithfulness was evaluated using an LLM-as-a-Judge approach, where a secondary model assessed whether the generated answer was fully supported by the retrieved context without introducing unsupported claims. Scores were averaged across the 20-question evaluation set.

**Analysis:** The high precision score confirms the effectiveness of the Arctic embedding model. The faithfulness score reflects the system's robust guardrails, particularly in refusing to answer questions (like JPM CEO names) not present in the provided context.

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## 6. Conclusion & Future Scope

The **Economic Intelligence Assistant** successfully demonstrates a production-ready RAG pipeline built natively on Snowflake.

**Future Enhancements:**

1. **Agentic Tool Use:** Enabling the model to perform SQL aggregations (e.g., "Count all banks in Texas") rather than relying solely on text chunks.

2. **Multi-Modal Support:** Integrating PDF parsing for complex financial annual reports.
3. **Expanded Corpus:** Including real-time FRED (Federal Reserve Economic Data) via API integration.